Classical Field Theory — Exercise Sheet 17

Forced and accidental massless directions

Solutions

Use the real scalar kinetic coefficient $1/2$ and the orbit–Hessian lemma of Lecture 17. Counts of massless modes below refer to the linearised field. A vacuum background need not belong to L^2; perturbation data may be Schwartz or in the earlier energy class.

Exercise 17.1 (The full orthogonal orbit). For $N \geq 3$, $\lambda, v > 0$, consider

$$V(u) = \frac{\lambda}{8}(|u|^2 - v^2)^2, \qquad u \in \mathbb{R}^N, \qquad u_0 = vn, \quad |n| = 1,$$

with the usual $O(N)$ action and canonical kinetic term. Find the vacuum set, the stabiliser at u_0, its dimension, and the orbit tangent. Derive the Hessian and diagonalise the linearised field using the radial and tangential projectors, without first rotating n to a coordinate axis. Identify which masses are symmetry-forced. Finally discuss $N = 1$ separately: does its broken discrete symmetry force a massless mode?

Solution. Nonnegativity shows that the minima are exactly $|u| = v$. The stabiliser fixes n and restricts to an arbitrary orthogonal transformation of $n^\perp$, so

$$\dim O(N) = \frac{N(N-1)}{2}, \qquad \dim H = \frac{(N-1)(N-2)}{2}, \qquad \dim O(N)/H = N - 1.$$

For every skew matrix T, $n \cdot Tn = 0$. Conversely, for $w \perp n$, $T = (wn^{\mathsf{T}} - nw^{\mathsf{T}})/v$ satisfies $Tu_0 = w$. Thus the orbit tangent is exactly $n^\perp$. Direct differentiation gives

$$\nabla V(u) = \frac{\lambda}{2}(|u|^2 - v^2)u, \qquad D^2V(u) = \frac{\lambda}{2}\big[(|u|^2 - v^2)I + 2uu^{\mathsf{T}}\big].$$

At u_0, this is $\lambda v^2 nn^{\mathsf{T}}$. For $P_\parallel = nn^{\mathsf{T}}$ and $P_\perp = I - P_\parallel$, the perturbation $h = nh_\parallel + h_\perp$ therefore obeys

$$(\Box + \lambda v^2)h_\parallel = 0, \qquad \Box h_\perp = 0.$$

The orthogonal decomposition preserves the canonical kinetic coefficient. All $N - 1$ zero modes are orbit directions, and no additional kernel remains. For $N = 1$, the vacuum set is the two points $\{-v, v\}$ and the stabiliser of either point in $O(1)$ is trivial. Both groups have dimension zero; there is no continuous orbit tangent. The Hessian is the positive scalar λv^2, so breaking this discrete symmetry gives no Goldstone mode.

Exercise 17.2 (A flat singlet still has a nonlinear restoring force). Add a real scalar z with its own canonical kinetic term and potential

$$\widetilde{V}(u, z) = \frac{\lambda}{8}(|u|^2 - v^2)^2 + z^4.$$

Let $O(N)$ act on u and trivially on z. Find the vacuum set, the Hessian kernel at $(vn, 0)$, and the subspace forced by the symmetry. Can the extra singlet direction be tangent to a curve of vacua? On the spatial torus, set $u = vn$ and take a homogeneous singlet $z(0) = A > 0$, $z_t(0) = 0$. Derive its nonlinear period as a definite integral and its dependence on A. Explain how a zero linearised mass is compatible with this nonlinear oscillation. These homogeneous data are not claimed to be L^2 data on $\mathbb{R}^d$.

Solution. The minima require $|u| = v$ and $z = 0$. At $(vn, 0)$ the Hessian and its kernel are

$$D^2\widetilde{V} = \begin{pmatrix} \lambda v^2 n n^{\mathsf{T}} & 0 \\ 0 & 0 \end{pmatrix}, \qquad \ker D^2\widetilde{V} = (n^{\perp} \times \{0\}) \oplus (\{0\} \times \mathbb{R}).$$

The orbit tangent is just $n^{\perp} \times \{0\}$, of dimension $N - 1$. Every curve of vacua has its singlet coordinate identically zero, so its tangent cannot be the extra singlet direction. That zero is accidental.

For the torus solution the singlet equation is $z_{tt} + 4z^3 = 0$. Multiplication by z_t gives

$$\frac{1}{2} z_t^2 + z^4 = A^4, \qquad |z_t| = \sqrt{2(A^4 - z^4)}.$$

The orbit goes between $-A$ and A; uniqueness for this smooth mechanical ODE and its time-reversal symmetry give four equal quarter-periods. Changing variable $z = As$ in one quarter gives

$$T_A = 4 \int_0^A \frac{\mathrm{d}z}{\sqrt{2(A^4 - z^4)}} = \frac{2\sqrt{2}}{A} \int_0^1 (1 - s^4)^{-1/2} \mathrm{d}s.$$

The integral is finite because its endpoint singularity is proportional to $(1 - s)^{-1/2}$. Thus the nonlinear restoring force gives a finite period at every nonzero amplitude, while $T_A \to \infty$ as $A \downarrow 0$. The zero Hessian captures the infinitesimal limit, not absence of the cubic restoring force or the presence of a vacuum curve.

Exercise 17.3 (An ineffective group factor belongs to the stabiliser). Take $G = SO(3) \times U(1)$ acting on a real triplet by $(R, e^{i\theta})u = Ru$. Use the radial potential of Problem 1 with $N = 3$ and vacuum $u_0 = v(2, -1, 2)/3$. Determine the full stabiliser and the differential of the orbit map $(c, \theta) \mapsto c \times u_0$ on $\mathbb{R}^3 \oplus \mathbb{R}$. Find a basis of its kernel, the number of broken directions and the Hessian nullity. Why would counting $\dim G$ or only counting rotations that fix u_0 give a wrong answer?

Solution. The rotation stabiliser consists of rotations around the line through $(2, -1, 2)$, hence is $SO(2)$. Every phase factor fixes every triplet. Therefore

$$H = SO(2)_{u_0} \times U(1), \qquad \dim G = 4, \quad \dim H = 2, \quad \dim G/H = 2.$$

The infinitesimal action is $c \times u_0$, independent of θ. Its kernel requires c parallel to $(2, -1, 2)$ and leaves θ arbitrary. A basis of the kernel in the four real parameters is

$$(2, -1, 2, 0), \qquad (0, 0, 0, 1).$$

Its rank is two. The Hessian is $\lambda u_0 u_0^{\mathsf{T}}$ of rank one, so its nullity is also two and matches the orbit tangent. Counting all four generators would include both unbroken parameters. Subtracting only the one-dimensional rotation stabiliser would still incorrectly count the ineffective phase as broken. The full stabiliser is what appears in the orbit–Hessian lemma.

Exercise 17.4 (Kinetic normalisation before diagonalising masses). For a real three-component perturbation h, consider the quadratic density

$$L = \frac{1}{2}\partial_\mu h^{\mathsf{T}} K \partial^\mu h - \frac{1}{2} h^{\mathsf{T}} M h, \qquad K = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 9 & 9 \\ 0 & 9 & 9 \end{pmatrix}.$$

The real phase data may be taken Schwartz. Verify the factorisation $K = S^{\mathsf{T}} S$ for

$$S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Derive the canonical momentum, and find the corresponding change of coordinates and momenta that preserves $\int \delta h_a \wedge \delta p_a$. Compute the physical squared masses and compare them with the raw eigenvalues of M. State the general eigenproblem and explain what happens to the kernel dimension under this kinetic normalisation.

Solution. Multiplying the given matrices gives $S^{\mathsf{T}} S = K$ and $\det S = -1$, so K is positive definite. The velocity derivative gives $p = K h_t$. For $Q = Sh$ and $P = S^{-\mathsf{T}} p$, the canonical potential is preserved:

$$p^{\mathsf{T}} \delta h = P^{\mathsf{T}} \delta Q.$$

Taking its negative exterior derivative proves preservation of the canonical form, including the momentum transformation's inverse transpose. Also $P = S h_t = Q_t$, so the kinetic term in Q is canonical. The transformed density is

$$L = \frac{1}{2}\partial_\mu Q^{\mathsf{T}} \partial^\mu Q - \frac{1}{2} Q^{\mathsf{T}} D Q, \qquad D = S^{-\mathsf{T}} M S^{-1} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{pmatrix}.$$

Thus its squared masses are $0, 4, 9$, whereas the nonzero two-by-two block of M itself has eigenvalues $0, 18$ in addition to its separate eigenvalue4. In general the field equation is $K \Box h + M h = 0$, giving

$$M w = m^2 K w$$

for a mode of squared mass m^2. An invertible congruence preserves the kernel dimension, even though it does not preserve the nonzero raw eigenvalues. Here the zero direction $(0, 1, -1)^T$ becomes $Q = e_1$. A symmetry-forced kernel therefore remains massless under normalisation, but physical nonzero masses must use the kinetic pairing as well as the potential Hessian.